

Word Embeddings and Semantic Geometry

DATA 443/643 · Week 1, note 1.4

After Stanford CS224N Lecture 2: distributional meaning, skip-gram softmax, then evaluation.

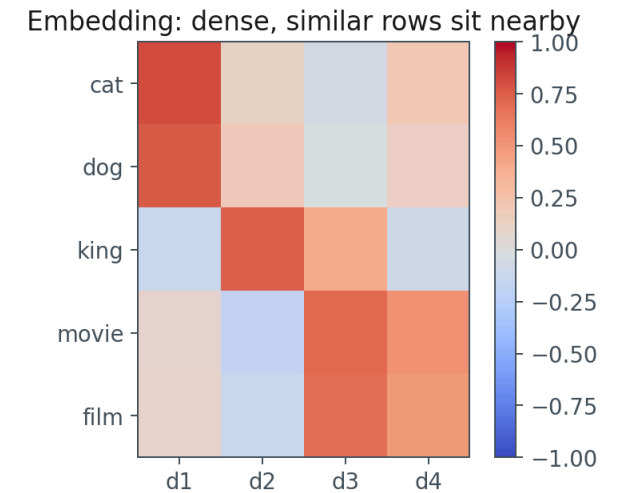
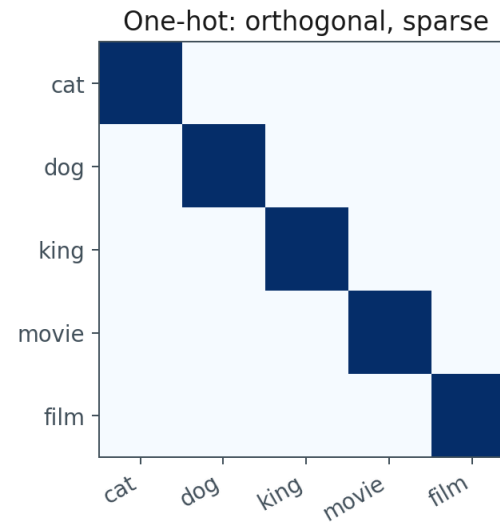
Lecture plan

- | | | |
|---|---|-------|
| 1 | One-hot is a bad geometry | 3 min |
| 2 | Distributional / distributed / embedding | 3 min |
| 3 | Skip-gram: window, softmax, lookup | 7 min |
| 4 | Intrinsic vs extrinsic, then bias as geometry | 5 min |

Key goal (CS224N): a word can be a dense vector; you can write $P(o \text{ given } c)$; projects report extrinsic numbers.

Problem with words as discrete symbols

- film and movie are orthogonal.
- No natural similarity for one-hot vectors.
- A new word has no coordinate.



Three words that are not synonyms

Distributional

Similar contexts, similar meaning.
Harris; Firth; Jurafsky Ch. 5.

Distributed

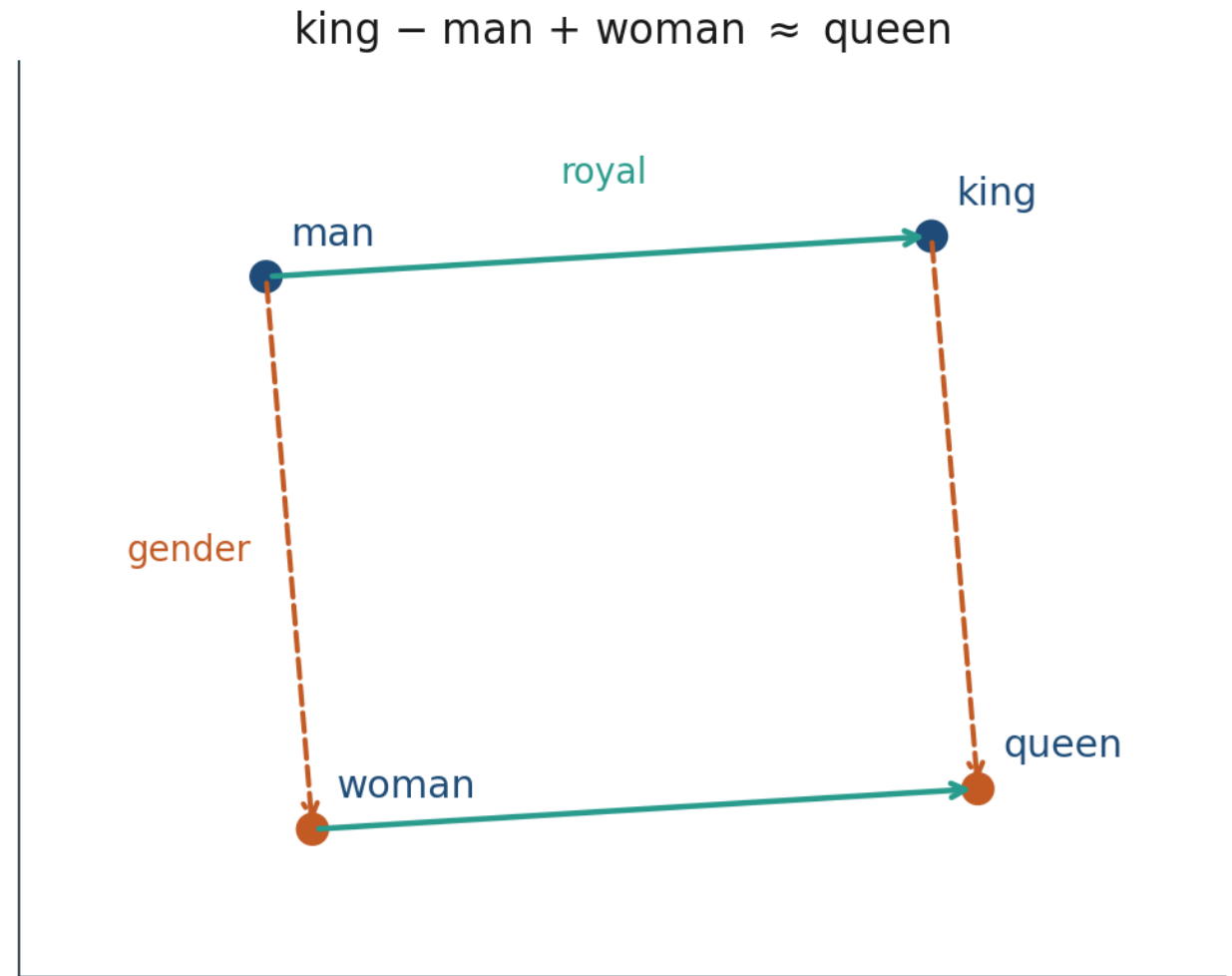
The code is a dense vector, not a one-hot.

Embedding

The lookup: `nn.Embedding`, a row of a V by d table.

Direction can mean something

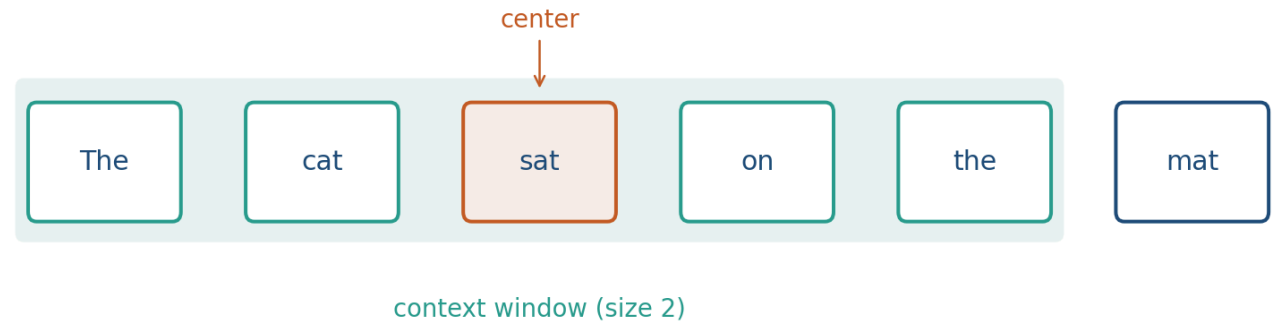
- $\text{king} - \text{man} + \text{woman} \approx \text{queen}$.
- Cosine: $u^T v / (\|u\| \|v\|)$.
- Nearest neighbors debug. They are not a project metric.



Skip-gram: from the center, predict a neighbor

- Fake task on a window.
- CBOW: from neighbors, predict the center.
- Logits $z_w = u_w^\top v_c$.

Skip-gram: predict each neighbor from the center word



Word2Vec prediction function

$$P(w_o \mid w_c) = \frac{\exp(u_o^\top v_c)}{\sum_w \exp(u_w^\top v_c)}$$

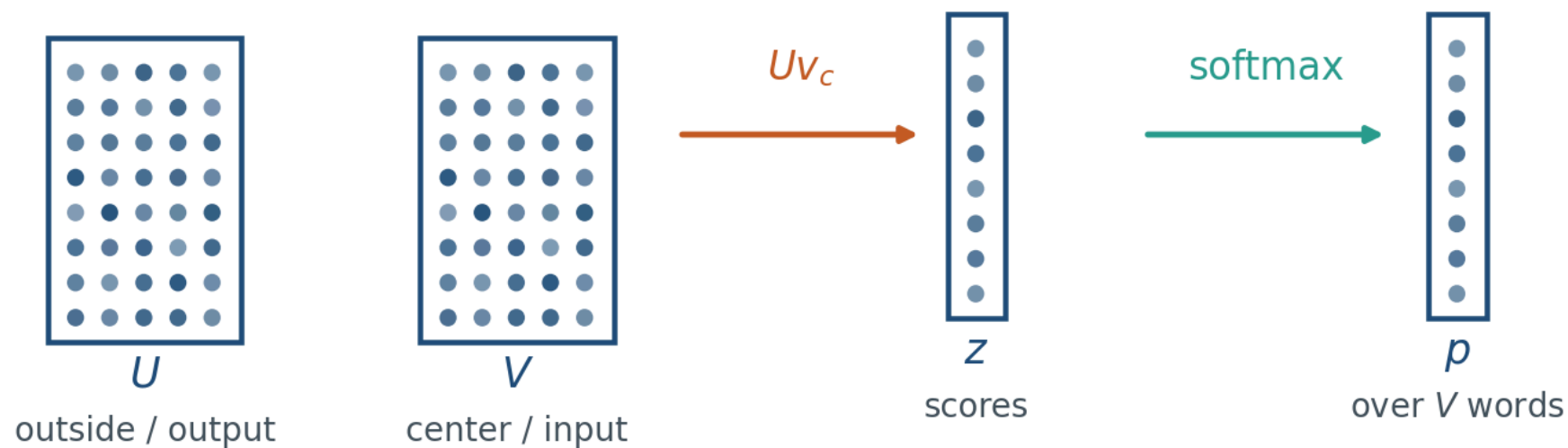
1. Dot product scores similarity of o and c.

2. Exp makes every score positive.

3. Normalize over the vocabulary. That is softmax.

Skip-gram is a table, a product, then a softmax

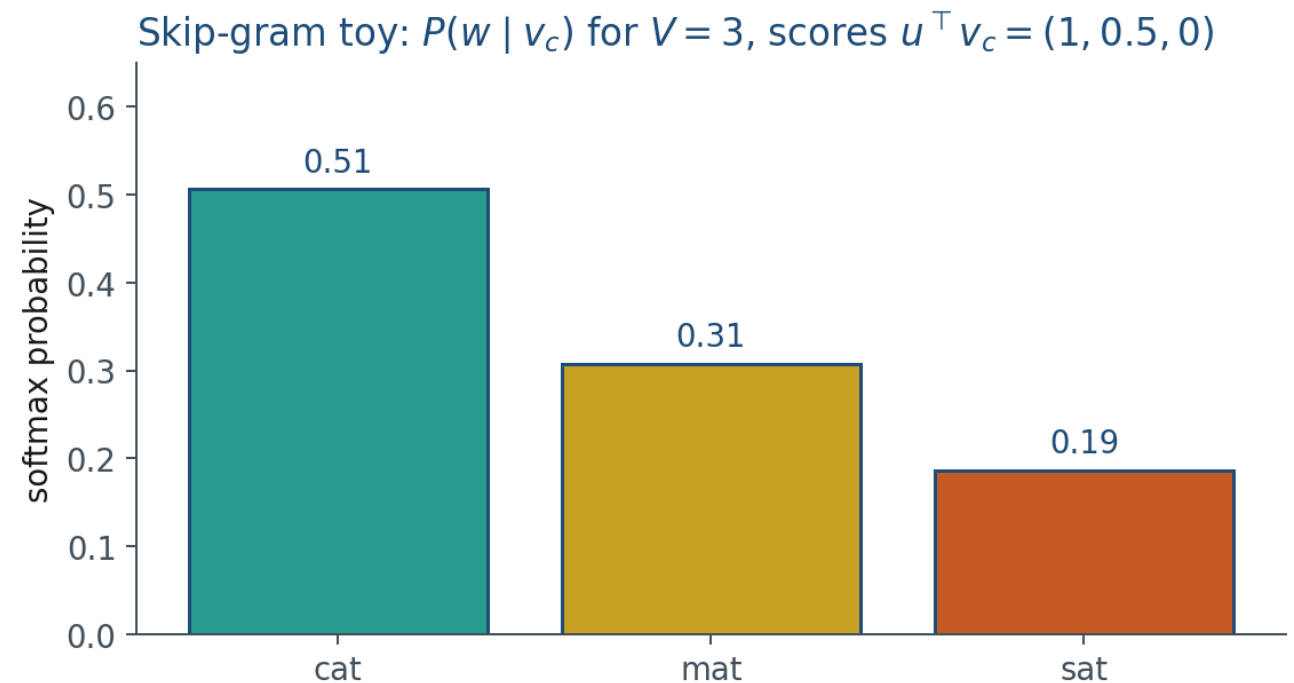
Skip-gram is a lookup, a dot product, then a softmax — not a mysterious net



Keep V (center rows). Throw away the softmax. A transformer keeps transforming those vectors.

A three-word softmax you can finish by hand

- Scores $(1, 0.5, 0)$ for cat, mat, sat.
- $p \approx (0.51, 0.31, 0.19)$.
- Mikolov 2013a §§1–3. Negative sampling is cited, not derived.



Counting vs. predicting

Count

Build a co-occurrence matrix, then factor it (LSA / GloVe). Classical vector semantics can start here.

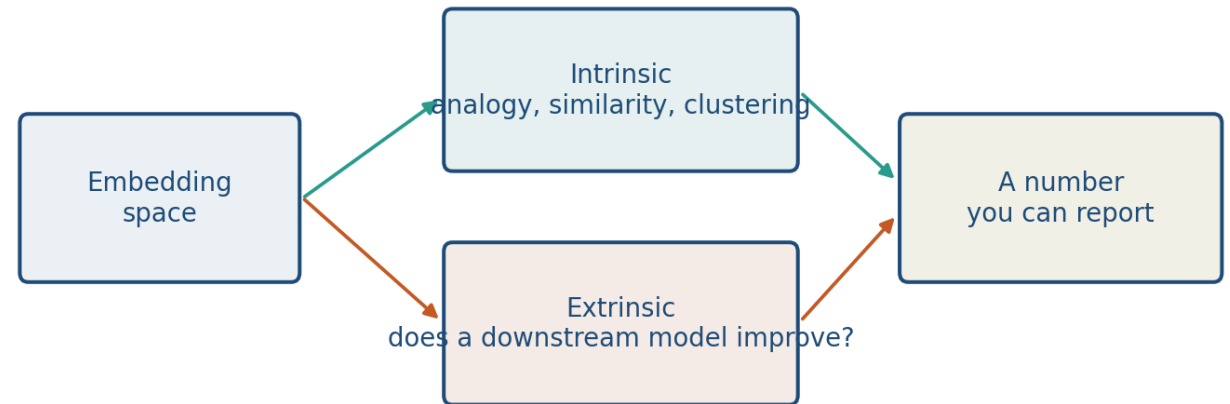
Predict

Skip-gram predicts context. Same distributional idea; same training loop as an LLM. No SVD homework this week.

How to evaluate word vectors

- Intrinsic: analogies, similarity, clustering.
- Extrinsic: does a downstream model improve?
- They need not agree. Projects report extrinsic.

Two ways to score embeddings. They need not agree.



Bias is also geometry

- Offset $o = \vec{he} - \vec{she}$.
- Score $v^T o / \|o\|$. Larger sits closer to he.
- Lab 1: constructed 2-D first, then the same probe on a real model later.

Geometry can encode stereotypes as well as meaning

